Autonomous Humanoid Robot to Combat Fires

HS-MEAR-275

Project Origin

- Climate change is causing more violent fires to break out, e.g. Jan 2025 Southern California Fires
- Firefighters are beginning to get overwhelmed and are lacking help
- Resources available to fight fires are not enough to fight large scale ones
- Forests are burning down putting more pressure on the planet
- Huge amounts of property and goods are being destroyed by fires

What and Why is it a Problem?

Large amount of fires being spread across California with limited firefighters to help fight it

- With limited funding being given to firefighters it is necessary to start helping them
- Budget this year for Los Angeles Fire Department was cut by 17.6 million dollars
- They were not able to recruit as many firefighters to help them
- Need technology to increase efficiency and decrease cost

- Large amount of property damage is being caused by these fires
- Insured losses estimated to be around 75 billion dollars lost
- Around 40,500 acres of land was burnt

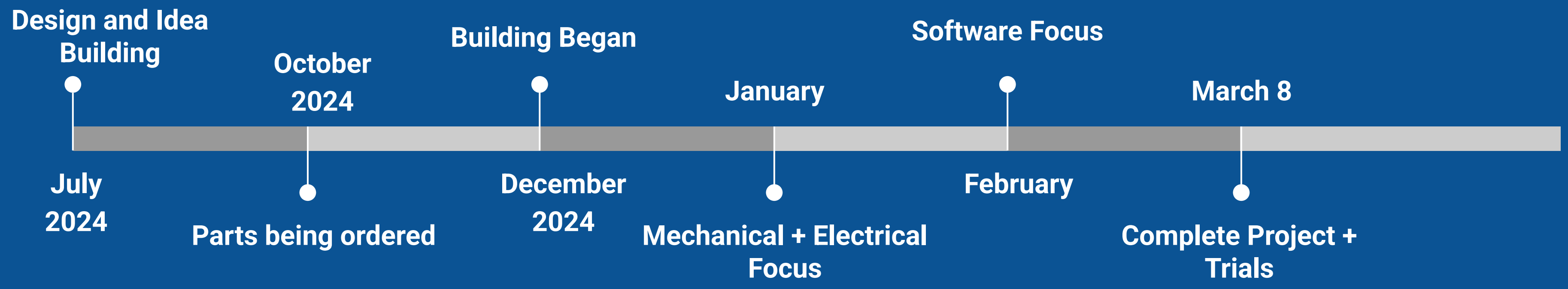
- Need for advancement in robotics automation
- Need to reduce the amount of money being put into Firefighting
- Need constant surveillance for danger spots in fires

Procedure

Created a humanoid robot that can walk, conduct environmental surveillance and fight fires by first creating a design of it in Onshape(Computer Aided Design Software). Next I began working on the mechanical piece of the robot fabricating pieces of the robot using different metal tools to create them as desired from used parts. Continued by beginning the assembly of the pieces and making minor edits to the design to take into account the weight of the batteries. After this the software was worked on. This was where a large amount of time was put in to create a script for walking and standing which would ensure the stability of the robot as it worked through terrain. Alongside this, object detection and specific environmental signs were ingrained into the A.I powering the camera. This was used for the robot to detect the fire hose, fires, etc. Once the software was perfected, data collection began where the robot was tested against an obstacle course to determine whether it could navigate through obstacles and reach its objective of dousing the fire promptly.

Solution Overview & Timeline

- Humanoid robot able to visually detect fire, walk close and point water hose to put it out
- Able to detect obstacles and maneuver past them
- Able to go over terrain that is rocky or uneven
- Cheap Robot that is mass manufacturable
- Easily transferable, can be compressed to be taken from Point A to Point B
- Able to conduct environmental surveillance to determine spots on the terrain where a fire may break out



Solution: Mechanical

Main Components

Legs	Adafruit Servo Board
Torso	12A 5V Battery Pack
Camera	5V Battery Pack
Raspberry Pi	14 Servos

Torso

- C-Channels and Plates to hold up electronics
- Holds Battery Packs and Camera

Arm

- 4 servos on each arm controlling → Claw, Elbow, 2 Shoulders
- Controlled by one 25kg servo, one 20kg servo and 2 regular servos
- Numerous degrees of freedom, capable of lifting 10-20 kg of load

Legs

- 3 Servos controlling → Hips side-side, Hip, Ankle side-side; 3 20kg on each
- Made out of connected C-Channels ensuring structural integrity
- Feet made out of wood braces connected together in a grid formation

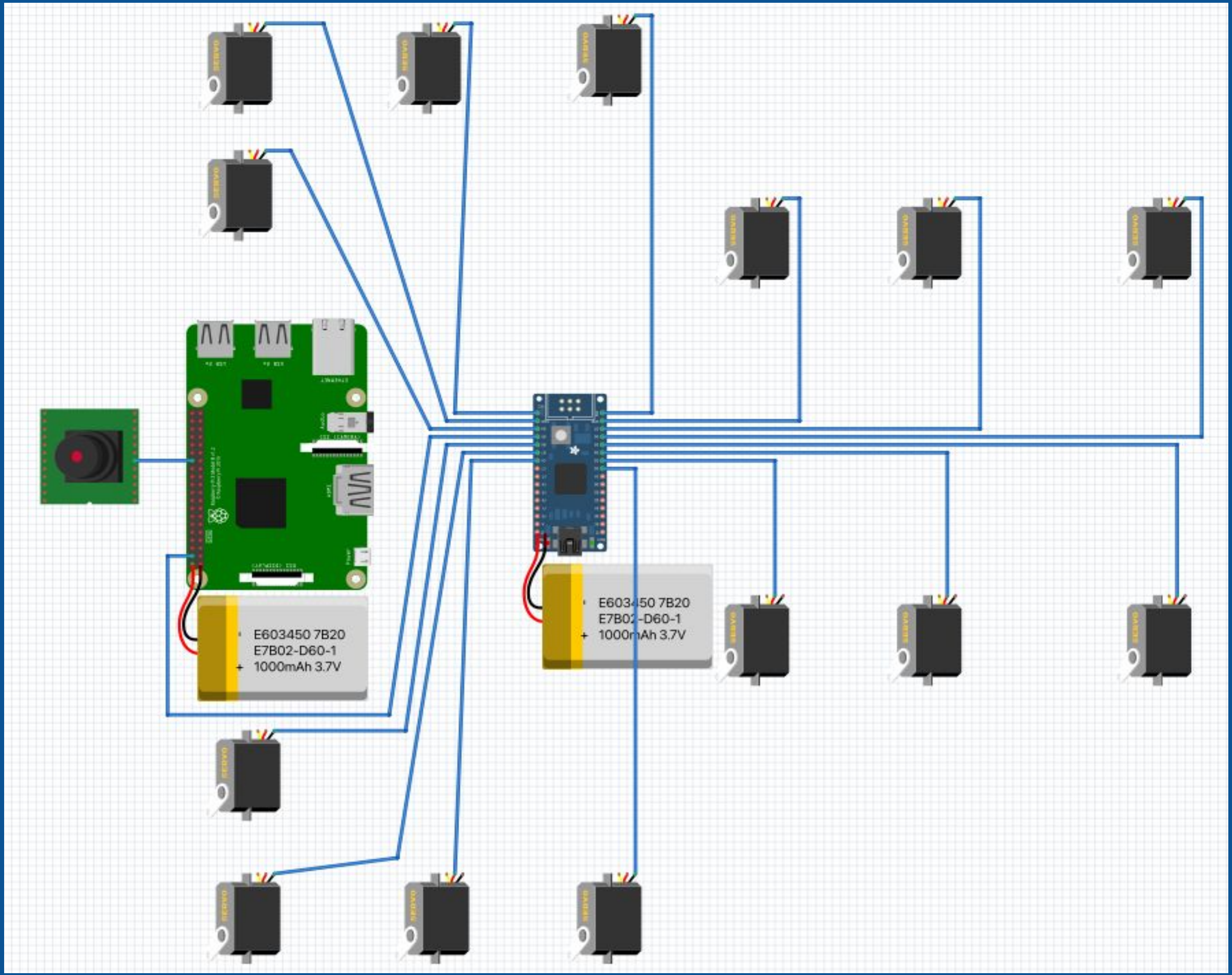
Solution: Software

```python
print("[INFO] start processing faces...")
imagePaths = list(paths.list_images("dataset"))
knownEncodings = []
knownNames = []

for (i, imagePath) in enumerate(imagePaths):

print(f"[INFO] processing image {i + 1}/{len(imagePaths)}")
    name = imagePath.split(os.path.sep`)[-2]

    image = cv2.imread(imagePath)
    rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    boxes = face_recognition.face_locations(rgb, model="hog")
    encodings = face_recognition.face_encodings(rgb, boxes)

    for encoding in encodings:
```

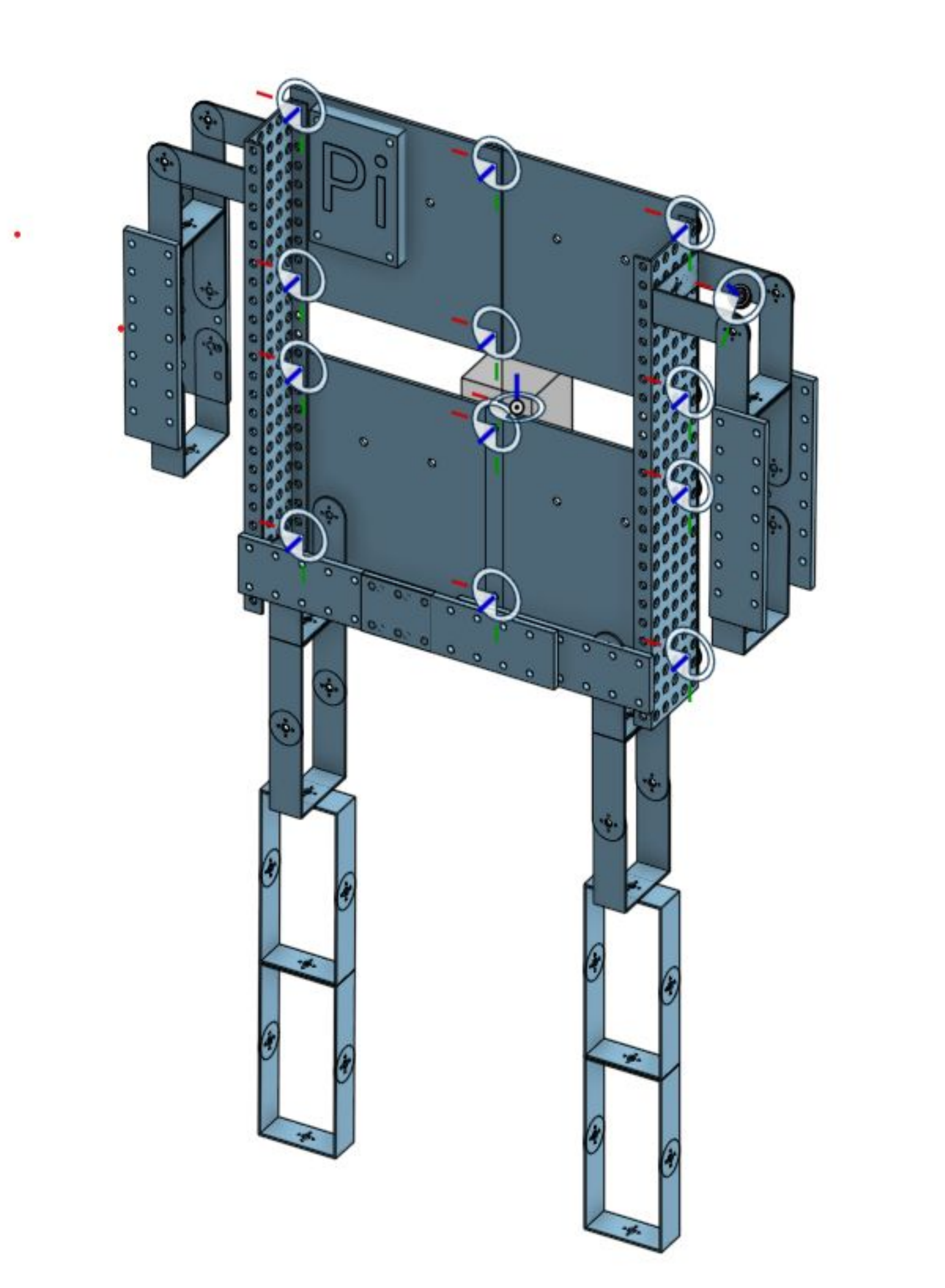
Overview

- Using Opencv for object detection; coding reactions based on camera input.
- Set walking script for repeated and consistent walking
- Using Thonny IDE
- Adafruit Servo Library
- Aiming for hose uses set constants to get accurate water shot

Circuit Schematic

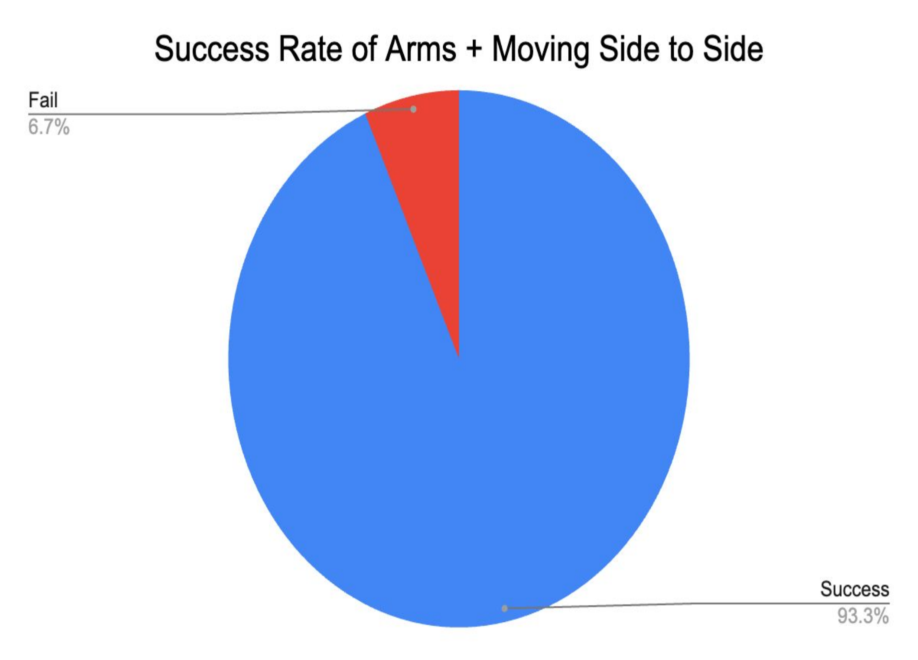


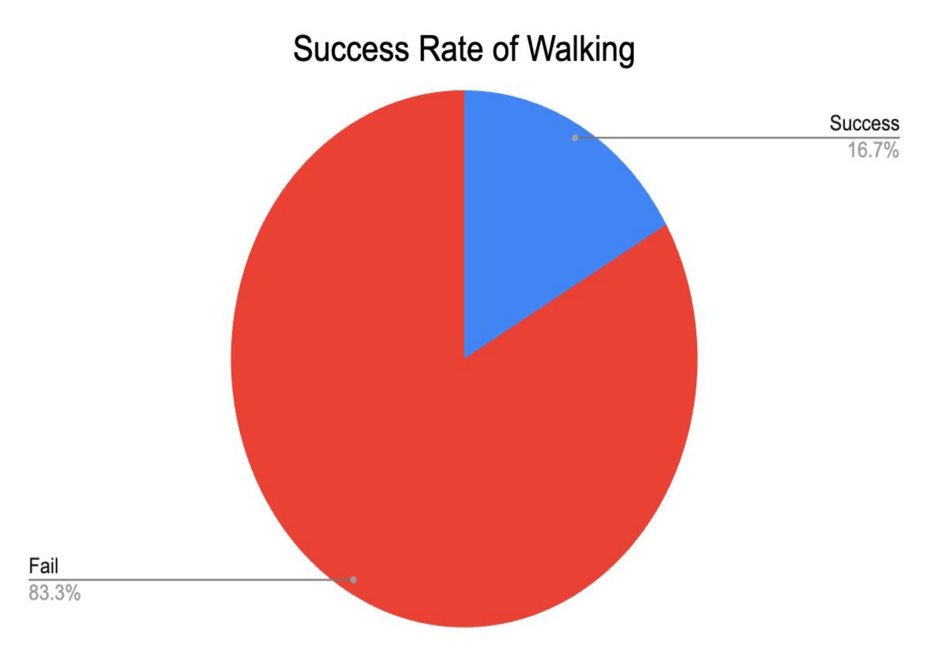
Design Schematic



Onshape Designed Schematic

Results





The success rate of auto alignment and dousing the fire was highly successful

The robot being able to walk to the target was inconsistent

Limitations and Next Steps

Limitations

- Has difficulty doing large degree turns limiting the robot
- Arms cannot go all the way back
- Camera cannot shift and only one is used limiting it to only look ahead
- Standing relies heavily on the ground being smooth
- No soft limits present resulting in it being able to move in positions where it could mechanically fail
- Has a hard time dealing with light directly in the camera → Sharp sun, flashing lights etc.

Next Steps

- Include turning feature to robot can turn its body while walking
- Higher FPS camera and the ability to process more data, with an A.I Hat
- Add in distance sensor to get estimates of where the robot is
- Increase the speed of walking
- Don’t have to charge to keep running; have the robot being powered by a solar panel
- Springs on legs to help cushion impact and ensure electronics and servos remain safe

Learnings and Alternate Uses

Learnings

- Set servos to 90 degrees before starting
- Design FIRST before building the robot
- Make sure all screws are properly tightened
- Work to secure wires and ensure your wiring is organized
- Always get more parts than you might need; Extra parts will always be used
- Take note of different positions of servos when coding software
- Have fun while building! :-)

Alternate Uses

- Manufacturing: Assisting in manufacturing plants to move large boxes and consumer goods
- Household help: Performing jobs that are tedious such as folding clothes, washing dishes etc.
- Military surveillance: Working with the military to help them scout the area and terrain
- Naval Help: Working on ships to fight fires and conduct inspections to ensure everything is working well

References

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FIRE

DEPT.


I´M NOT ALLOWED IN HELL
FIRE
DEPT
I PUT THE FIRE OUT